Sri Lanka Institute of Information Technology

BSc (Hons) in Information Technology Specializing in Data Science

IT3021 – Data Warehousing and Business Intelligence

Year 3 Semester 1

Assignment 2

Moderage I. U. - IT23834606

Contents

Step 1: Data Source for the Assignment 2

Description of the data source:

Dataset Name: Indian Store Data

Source: https://www.kaggle.com/datasets/abuhumzakhan/store-data

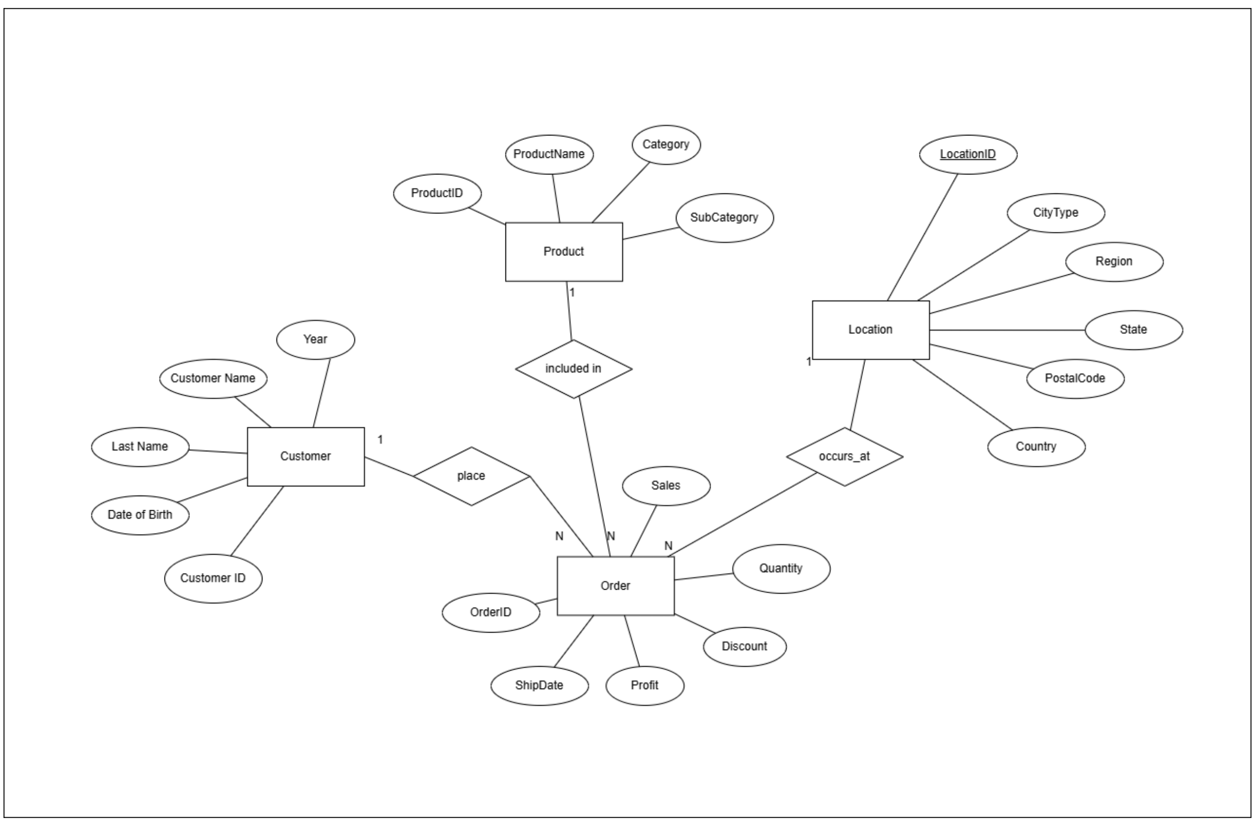
Data Period: The dataset spans over multiple years of sales transactions.

Data Format: SQL Database files, CSV files, Excel files

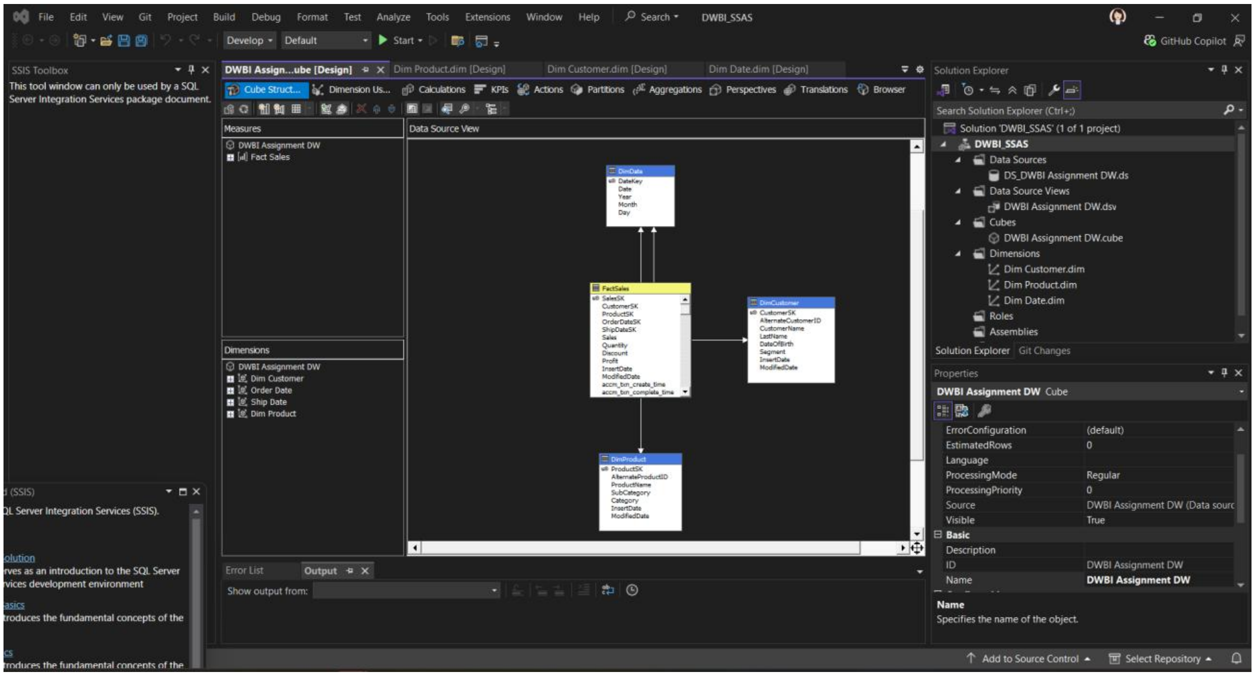
Description:

The Indian Store Data Database is a sample dataset that simulates real-world retail business operations. It contains approximately 100,000 records, making it ideal for data warehousing, ETL, SSAS cube modeling, and business intelligence (BI) reporting. This dataset follows an **OLTP structure**, making it suitable for analysis and data warehouse design.

ER Diagram:



Overall System Diagram:



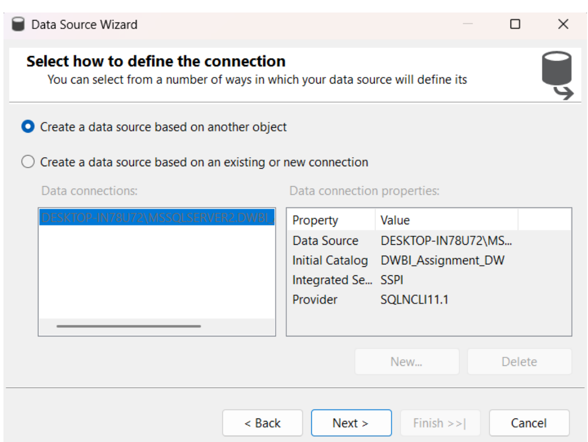
Step 2: SSAS Cube Implementation

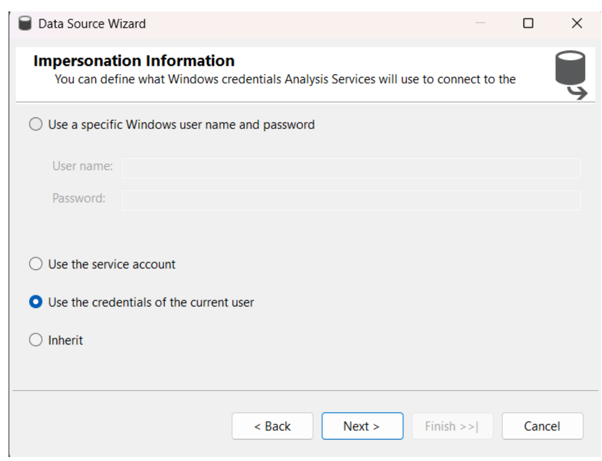
1. Creating the SSAS project:

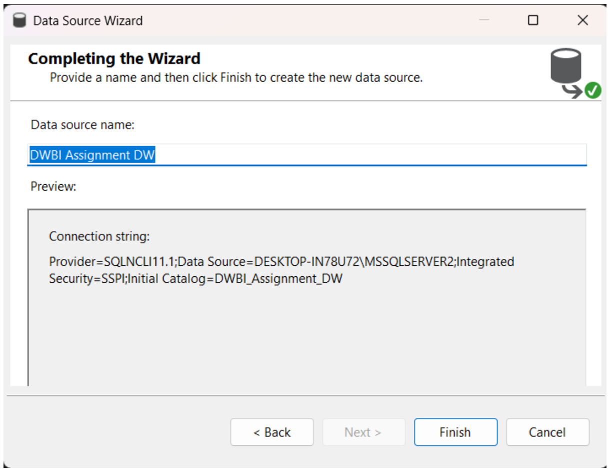
- Opened Visual Studio.
- Created a new project:
  - Project Type: Analysis Services Multidimensional and Data Mining Project
  - Project Name: DWBI Assignment DW
- Clicked OK to create the SSAS project workspace.

2. Setting up Data Source:

- Right-clicked on **Data Sources** → **New Data Source**.
- Selected existing **Data Warehouse** database as source.
- Chose the connection:
  - Database: DWBI_Assignment_DW
- Selected Use the service account.
- Completed the Data Source wizard.

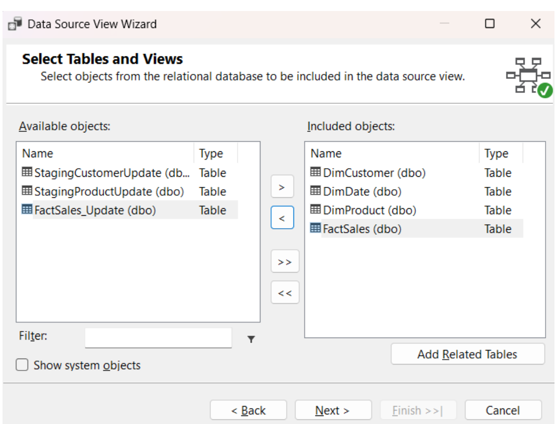


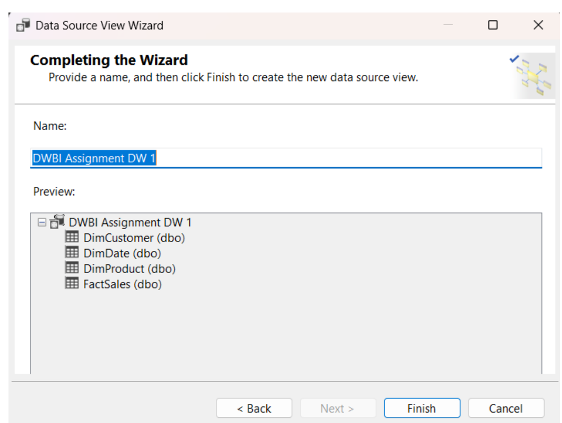




3. Setting up Data Source View (DSV)

- Right-clicked on **Data Source Views** → **New Data Source View**.
- Added relevant tables:
  - **Fact Table** (FactSales)
  - **Dimension Tables** (DimCustomer, DimDate, DimProduct)
- Verified relationships were automatically detected (Fact Table foreign keys to Dimension primary keys).

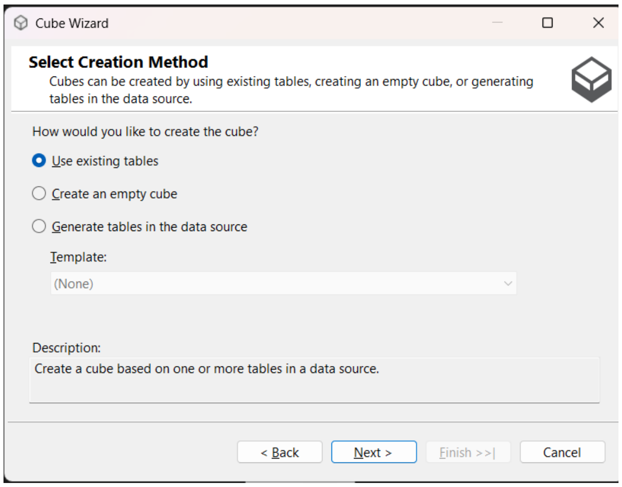


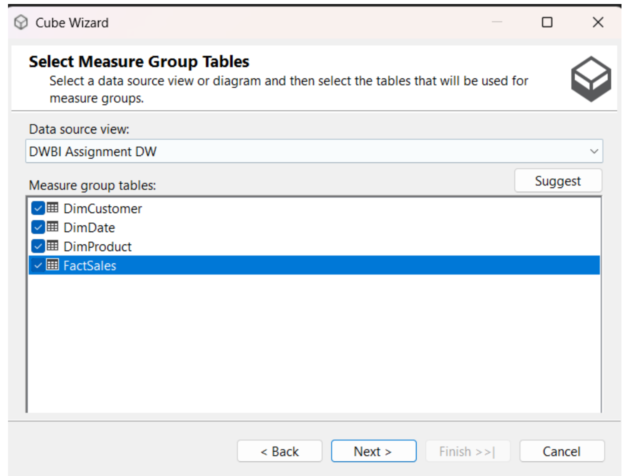


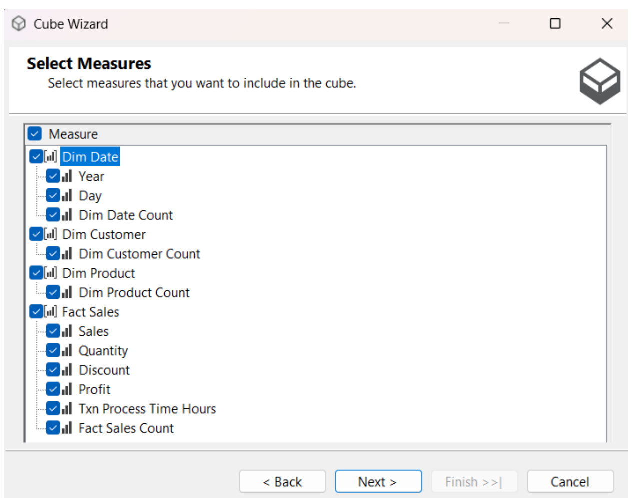
4. Creating the Cube

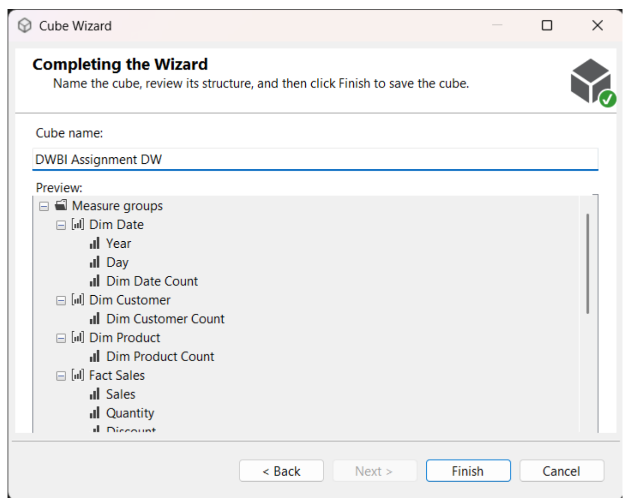
- Right-clicked on **Cubes** → **New Cube** → **Cube Wizard**.
- Selected **Use Existing Tables**.
- Chose the **Fact Table** (FactSales).

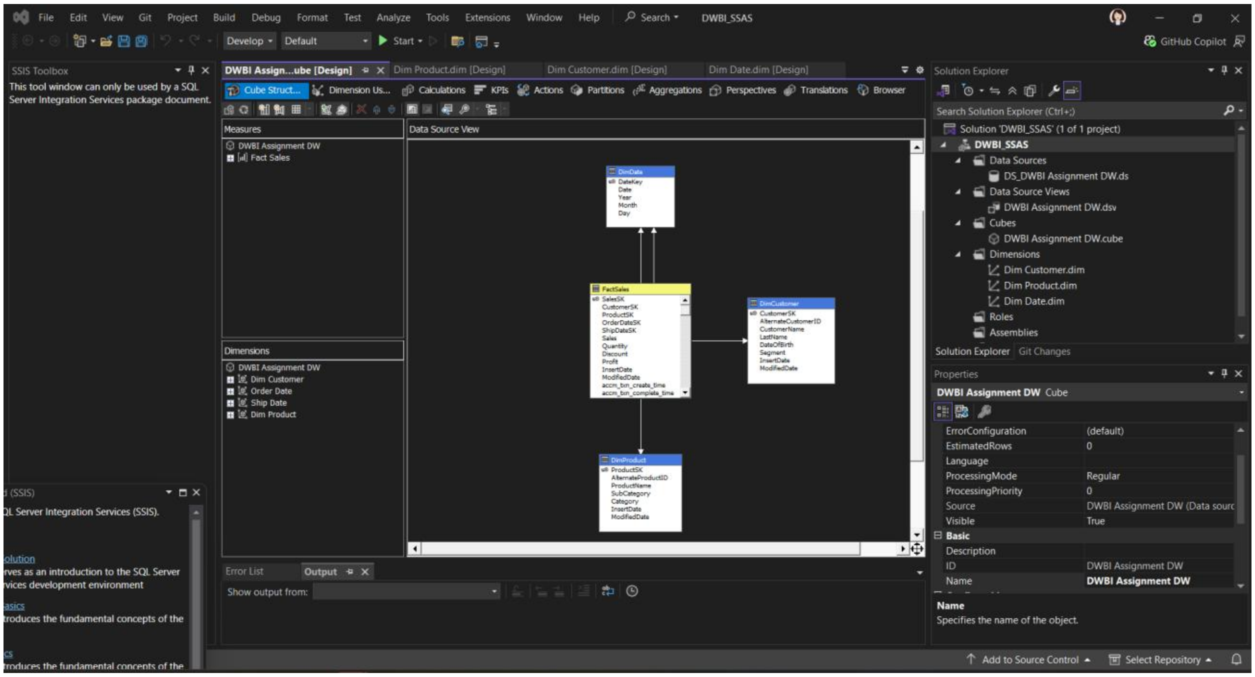
- Selected measures automatically detected by the wizard.
- Added related dimensions (already connected in DSV).
- Completed the Cube Wizard.









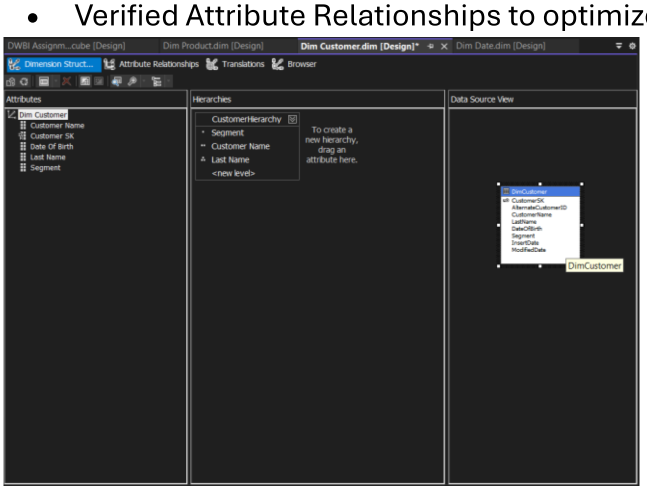


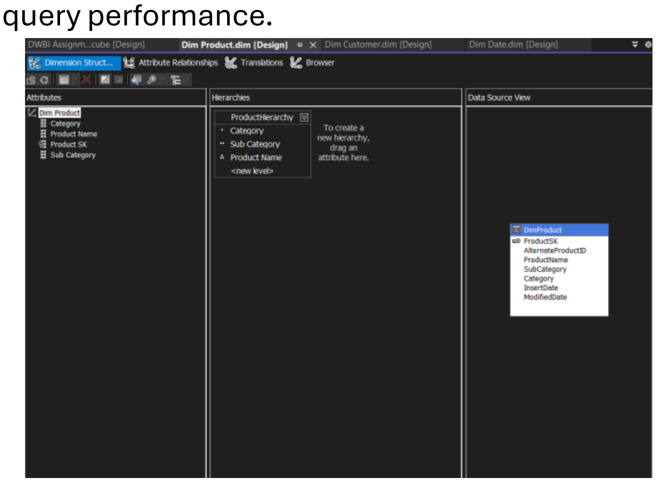
5. Designing Measures and Dimensions

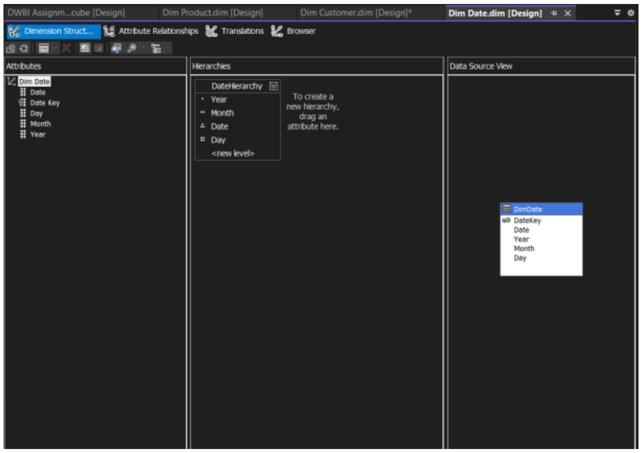
- Verified that Measures were correctly created under the Measure Group (FactSales).
- Configured dimension properties:

6. Implementing Hierarchy

- Opened Dimension Designer (**DimDate.dimension, DimCustomer.dimension**).
- Created **Hierarchies**:
    - Ex: Dragged attributes, **Year** → **Month** → **Date** to form a hierarchy.
- Verified Attribute Relationships to optimize query performance.



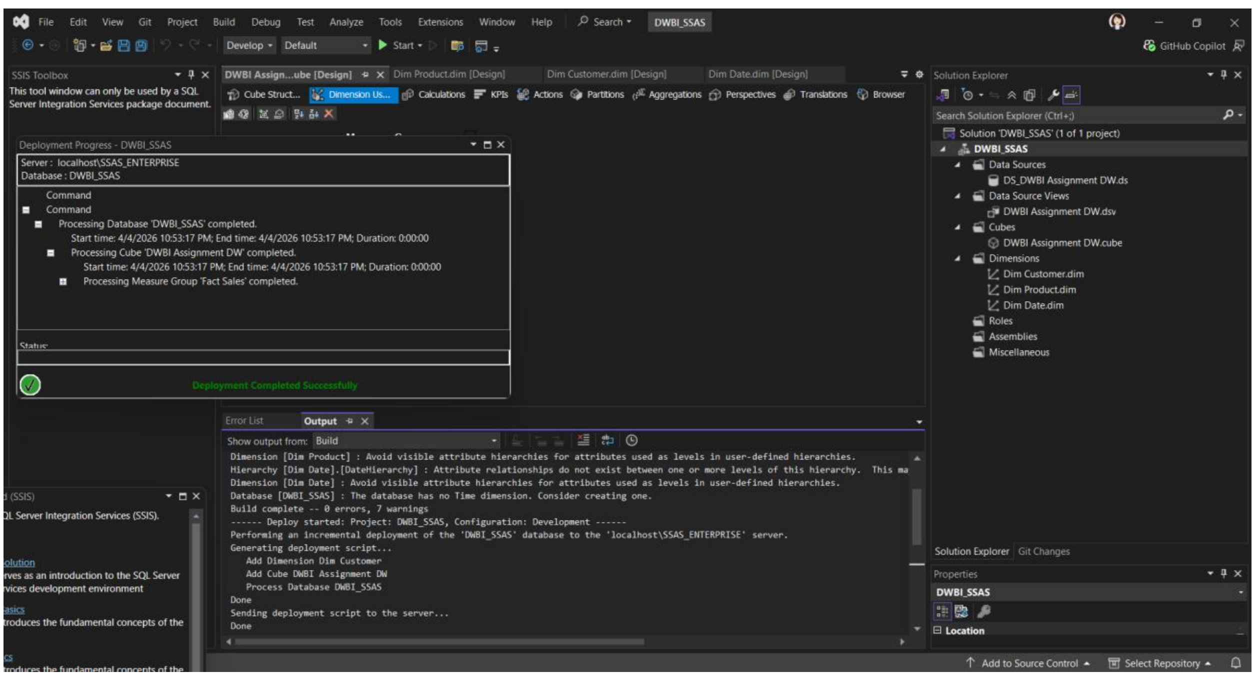




7. Deploying and Processing the Cube

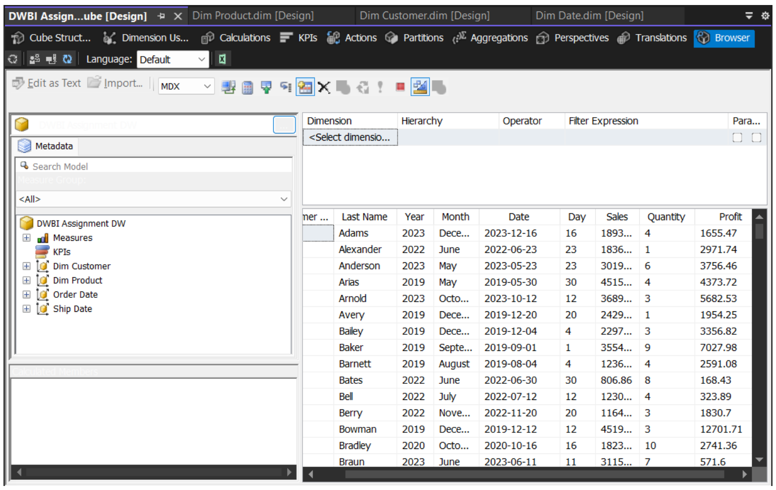
- Right-clicked the project → **Properties**.
- Set **Deployment Target Server**.
- Clicked **Deploy** to deploy cube to SSAS server.
- After successful deployment, **processed** the cube:
    - Right-click Cube → **Process** → Start.

Verified cube data loaded successfully.



8. Testing Cube

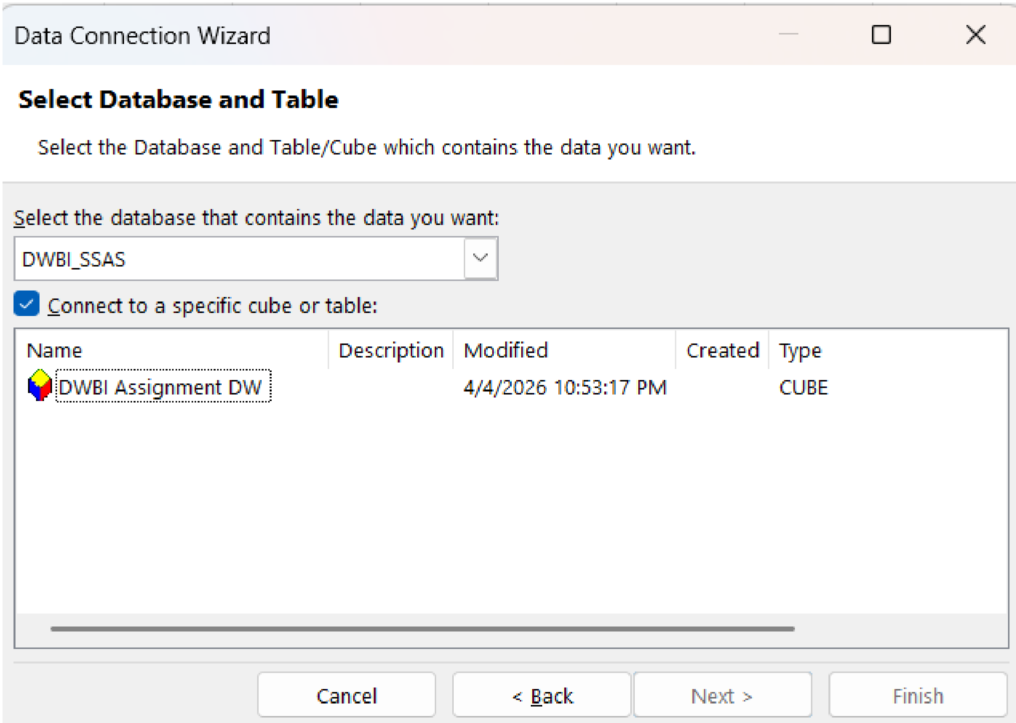
- Opened **Cube Browser** in Visual Studio.
- Dragged measures and dimensions into the browser to test:
    - Confirmed correct aggregation of measures.
    - Tested drill-down through the hierarchy (Year → Month → Day).



Step 3: Demonstration of OLAP operation

1. Connection to SSAS Cube

- Opened Microsoft Excel.
- Navigated to **Data → Get Data → From Database → From Analysis Services**.
- Entered server name
- Selected database DWBI_Assignment_DW and DWBI Assignment DW
- Created a PivotTable connected to the cube.

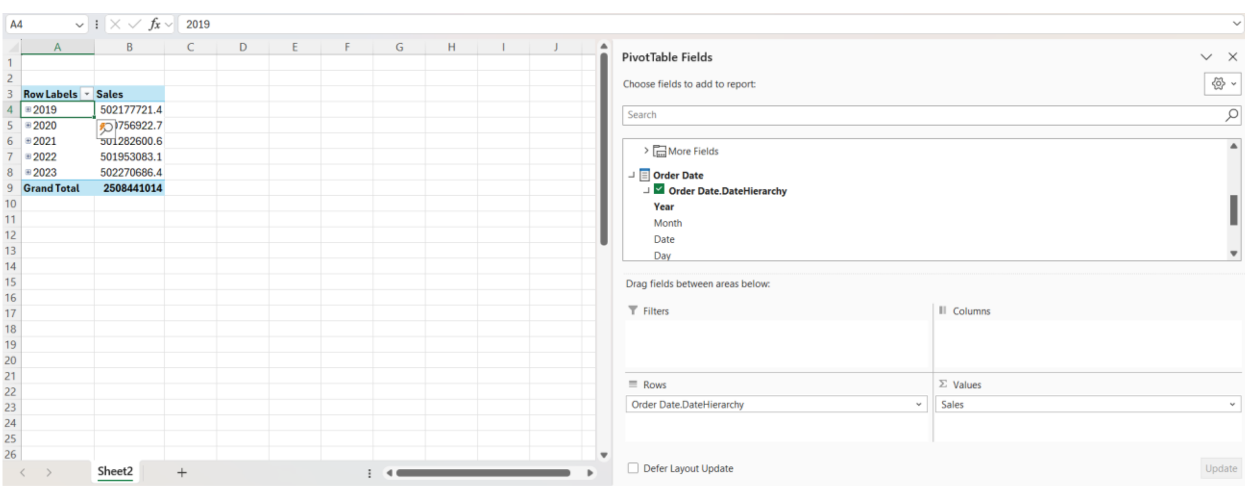


2. Roll-up Operation

- Dragged:
  - Date hierarchy → Rows
  - Sales → Values
- Collapsed hierarchy to show:

Year level summary

This demonstrates aggregation of data at a higher level.

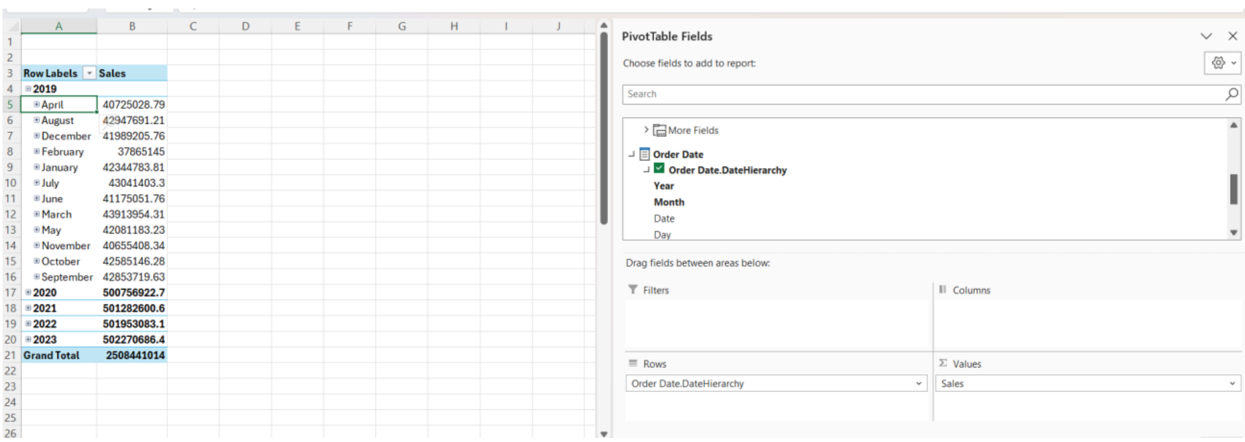


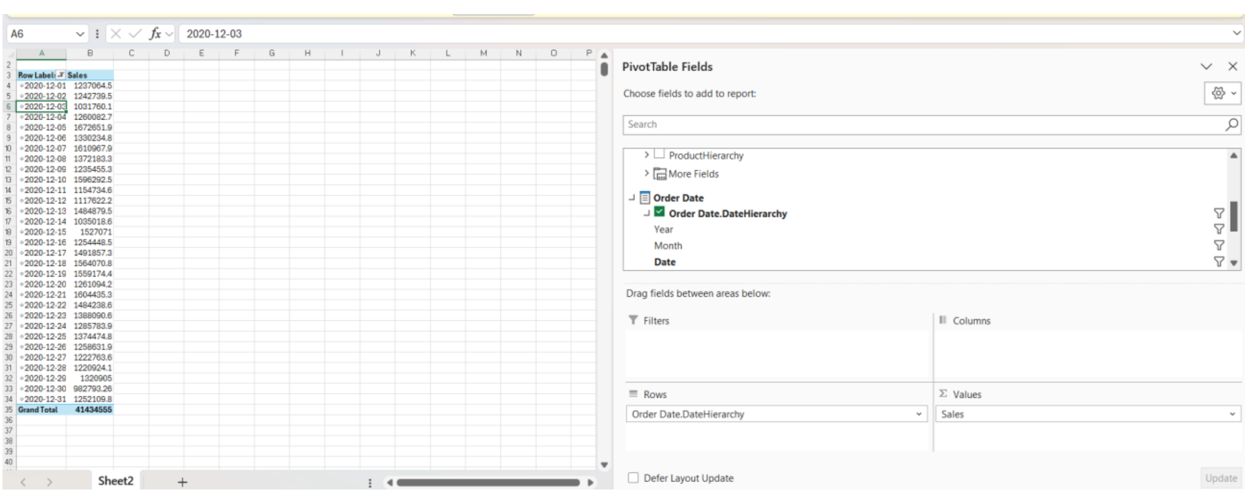
3. Drill-down Operation

- Expanded hierarchy:

Year → Month → Day

This shows detailed data at lower levels.





4. Slice Operation

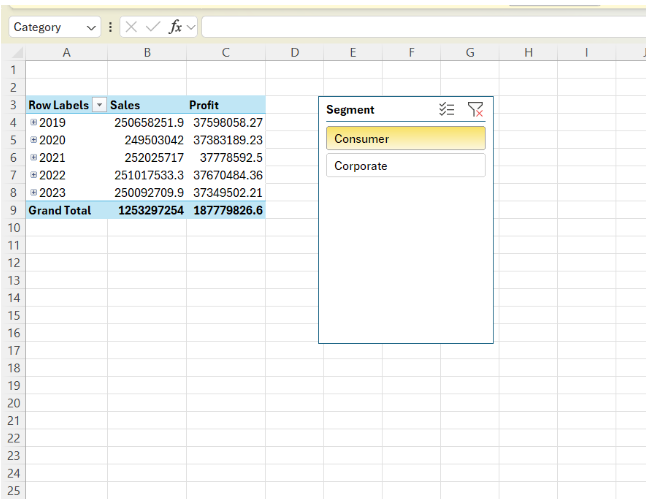
- Inserted slicer:

- Field: Segment

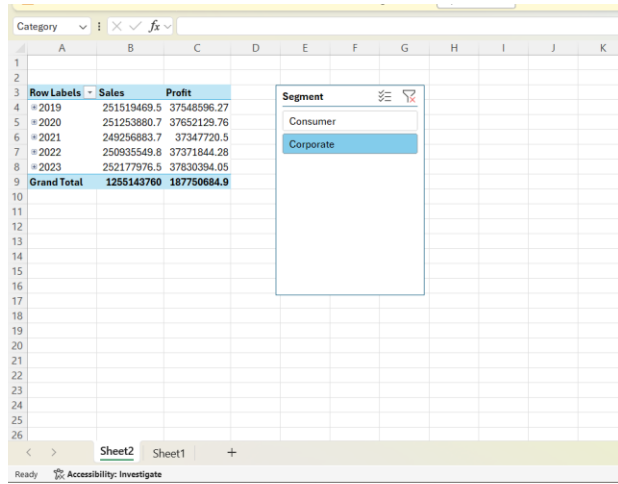
- Selected one value:

Consumer

PivotTable filtered to show only selected segment



Row Labels	Sales	Profit
2019	250658251.9	37598058.27
2020	249503042	37383189.23
2021	252025717	37778592.5
2022	251017533.3	37670484.36
2023	250092709.9	37349502.21
Grand Total	1253297254	187779826.6



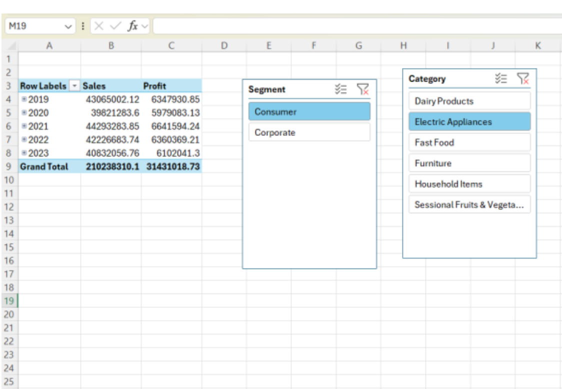
Row Labels	Sales	Profit
2019	251519469.5	37548596.27
2020	251253880.7	37652129.76
2021	249256883.7	37347720.5
2022	250935549.8	37371844.28
2023	252177976.5	37830394.05
Grand Total	1255143760	187750684.9

5. Dice Operation

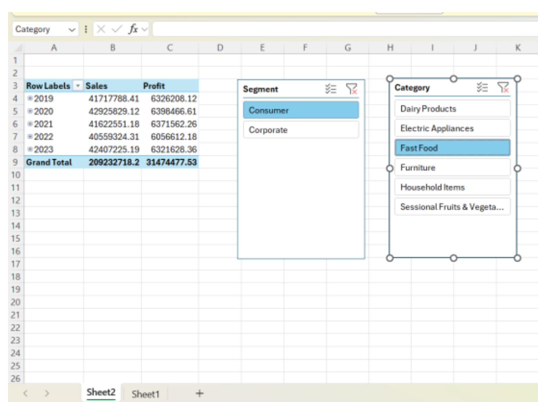
- Inserted multiple slicers:
  - Segment
  - Product Category
- Applied filters:

Consumer + Electronics

Displays a subset of data based on multiple conditions.



Row Labels	Sales	Profit
2019	43065002.12	6347930.85
2020	39821283.6	5979083.13
2021	44293283.85	6641594.24
2022	42226683.74	6360369.21
2023	40832056.76	6102041.3
Grand Total	210238310.1	31431018.73

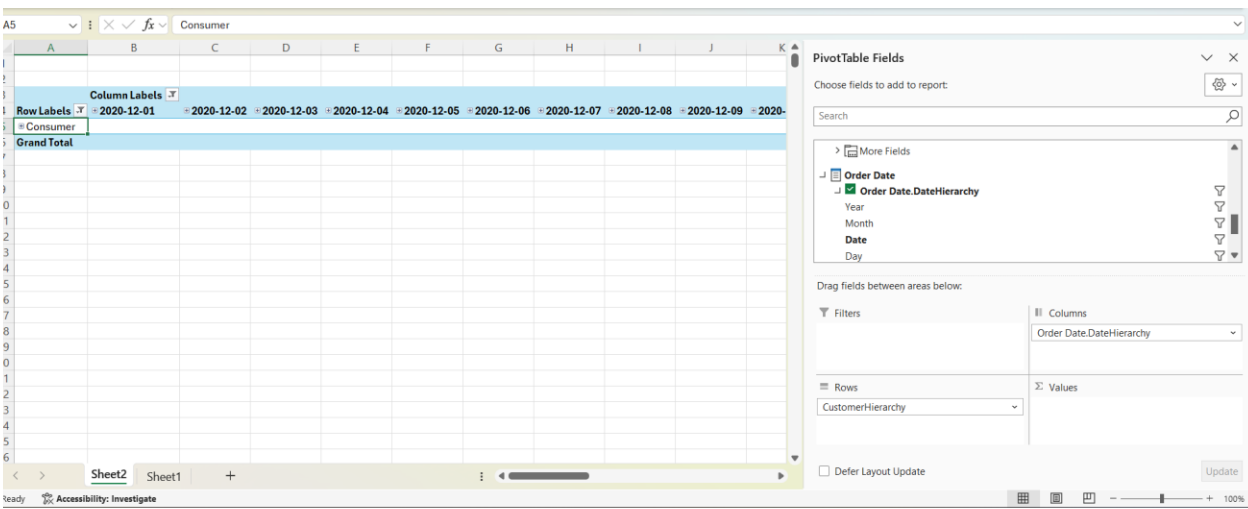


Row Labels	Sales	Profit
2019	41717788.41	6326208.12
2020	42925829.12	6398466.61
2021	41622551.18	6371562.26
2022	40559324.31	6056612.18
2023	42407225.19	6321628.36
Grand Total	209232718.2	31474477.53

6. Pivot Operation

- Rearranged fields:
  - Rows → Segment
  - Columns → Year
- Swapped:
  - Rows → Year
  - Columns → Segment

This changes the data perspective without altering the data.



Step 4: Power BI Reports

Steps:

- Open Power BI Desktop.

Connected to the OLAP Cube : **Home → Get Data → More... → Database → SQL Server Analysis Services**.

- Build Relationships between tables (use Model view).
- Save the Power BI file.

1. Report 1: Matrix Visual (Tabular Data with Row & Column Groupings)

Goal:

Create a report that displays detailed tabular data by organizing information into row and column groupings, allowing users to easily compare values across different categories.

Steps:

1. In Power BI Desktop, navigated to the **Report view** where visualizations are created and arranged.
2. Selected the **Matrix visual** from the Visualizations pane, which allows data to be displayed in a structured tabular format with hierarchical grouping.
3. Dragged a dimension, **Product Category (from DimProduct)**, into the **Rows** field well to categorize the data vertically.
4. Dragged another dimension, **Customer Segment (from DimCustomer)**, into the **Columns** field well to group the data horizontally.
5. Dragged a measure, **Sales (from FactSales)**, into the **Values** field well to display the aggregated numerical data within the matrix.
6. Applied formatting options to enhance readability and presentation:
    - Enabled **row and column subtotals** to display summarized values.
    - Adjusted **font size, alignment, and colors** for better clarity.
    - Styled the matrix with appropriate **gridlines and background colors**.

Drill on Rows

Customer Sales by Order Date

Segment	2019	2020	2021	2022	2023	Total
⊞ Consumer						
Sales	250,658.25K	249,503.04K	252,025.72K	251,017.53K	250,092.71K	**1,253,297.25K**
Quantity	55420	54637	54993	54736	55097	**274883**
Profit	37,598,058.27	37,383,189.23	37,778,592.50	37,670,484.36	37,349,502.21	**187,779,826.57**
⊞ Corporate						
Sales	251,519.47K	251,253.88K	249,256.88K	250,935.55K	252,177.98K	**1,255,143.76K**
Quantity	54811	54682	54950	54935	55115	**274493**
Profit	37,548,596.27	37,652,129.76	37,347,720.50	37,371,844.28	37,830,394.05	**187,750,684.86**
Sales	**502,177.72K**	**500,756.92K**	**501,282.60K**	**501,953.08K**	**502,270.69K**	**2,508,441.01K**
Quantity	**110231**	**109319**	**109943**	**109671**	**110212**	**549376**
Profit	**75,146,654.54**	**75,035,318.99**	**75,126,313.00**	**75,042,328.64**	**75,179,896.26**	**375,530,511.43**

2. Report 2: Slicers (Cascading Filters + Multiple Visuals)

Goal:

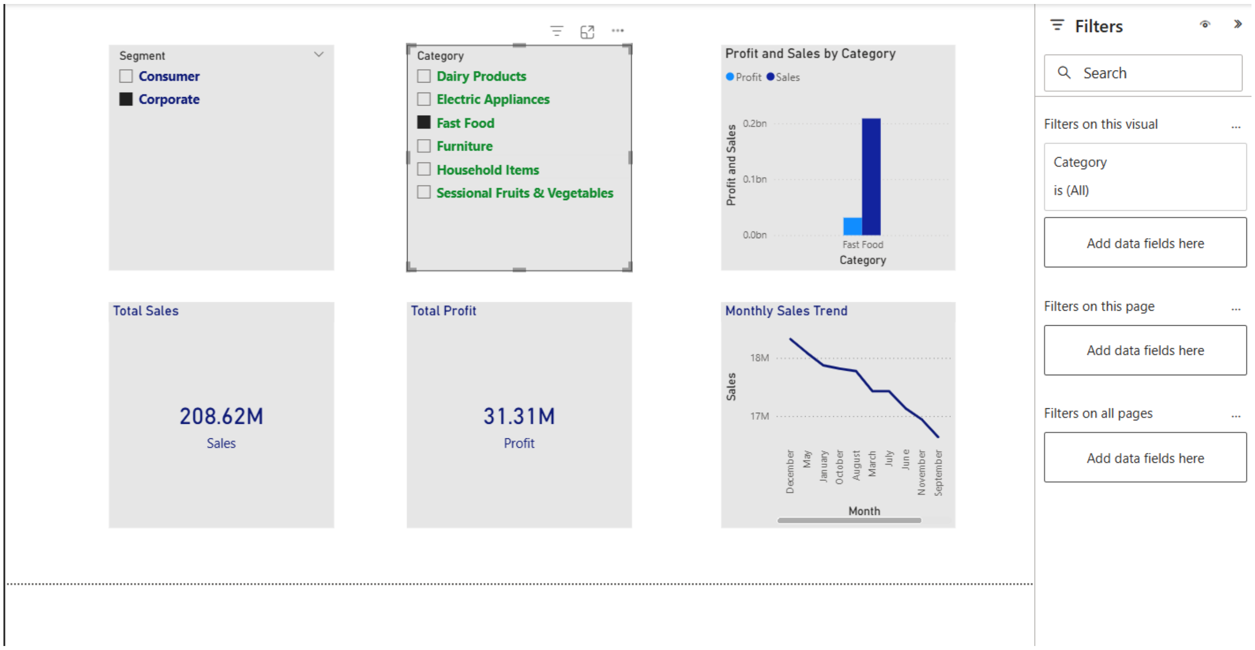
Create an interactive report using multiple slicers where one slicer dynamically filters another (cascading effect), along with multiple visuals to present insights.

Steps:

1. Inserted two **Slicer visuals** from the Visualizations pane.
2. Configured the first slicer by dragging the **Customer Segment (from DimCustomer)** field into it. This acts as the primary filter.
3. Configured the second slicer by dragging the **Product Category (from DimProduct)** field. This slicer dynamically updates based on the selection in the first slicer.
4. Verified cascading behavior, where selecting a value in the first slicer (e.g., a specific segment) filters the available options in the second slicer.
5. Added multiple visuals to represent the filtered data:
   - **Column Chart** showing Sales by Product Category.

- **Line Chart** showing Sales trends over time.

6. Applied formatting:
   - Used consistent **color themes** across visuals.
   - Adjusted **titles, labels, and legends**.
   - Positioned slicers and visuals neatly for better user interaction.



3. Report 3: Drill-Down Report (Hierarchical Data Exploration)

Goal:

Allow users to explore data at multiple levels of detail by drilling down through a hierarchy.

Steps:

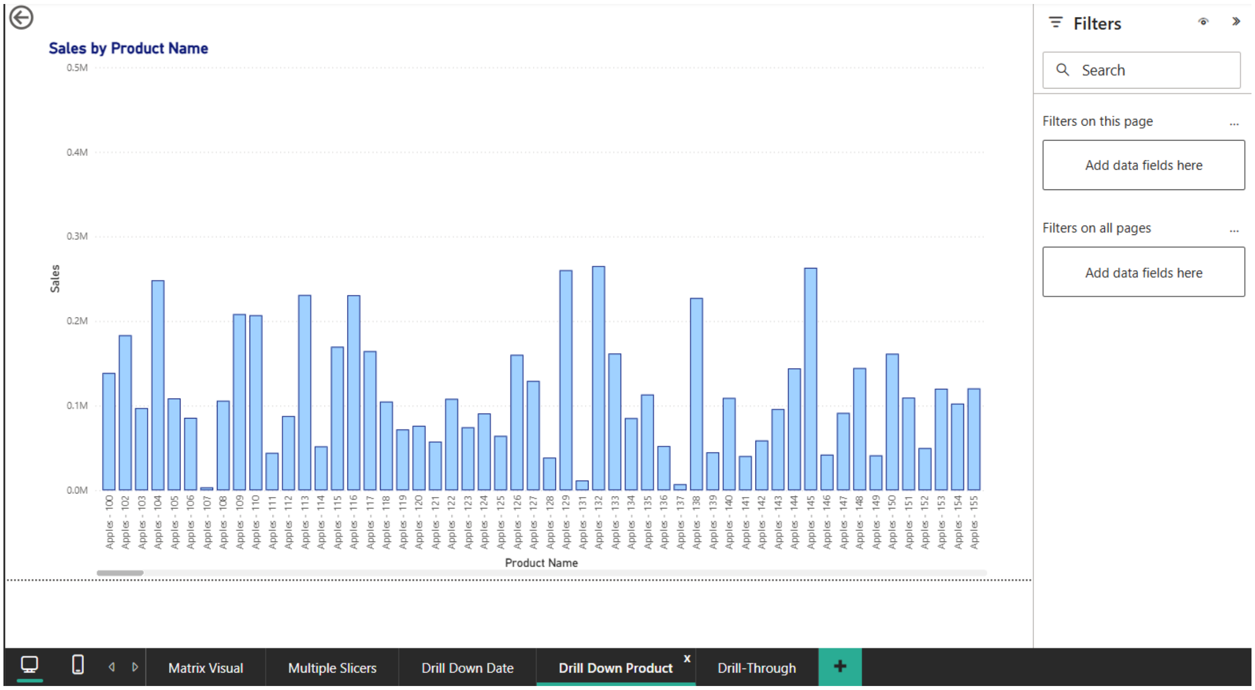
1. Inserted a **Clustered Column Chart** from the Visualizations pane.
2. Created a hierarchy by dragging multiple fields into the Axis area in the correct order:
   - **Year → Month → Day (from DimDate)**
3. Dragged a measure, **Sales (from FactSales)**, into the **Values** field.

4. Enabled drill-down functionality by clicking the **drill-down icon (↓)** located at the top of the visual.

5. Tested the drill-down feature by clicking on a specific year, which expanded the view to quarters, and further to months.

6. Applied formatting:

    - Added a clear **title**.
    - Adjusted **axis labels and data colors**.
    - Ensured readability and consistency.



4. Report 4: Drill-Through Report (Right-Click to Detailed Page)

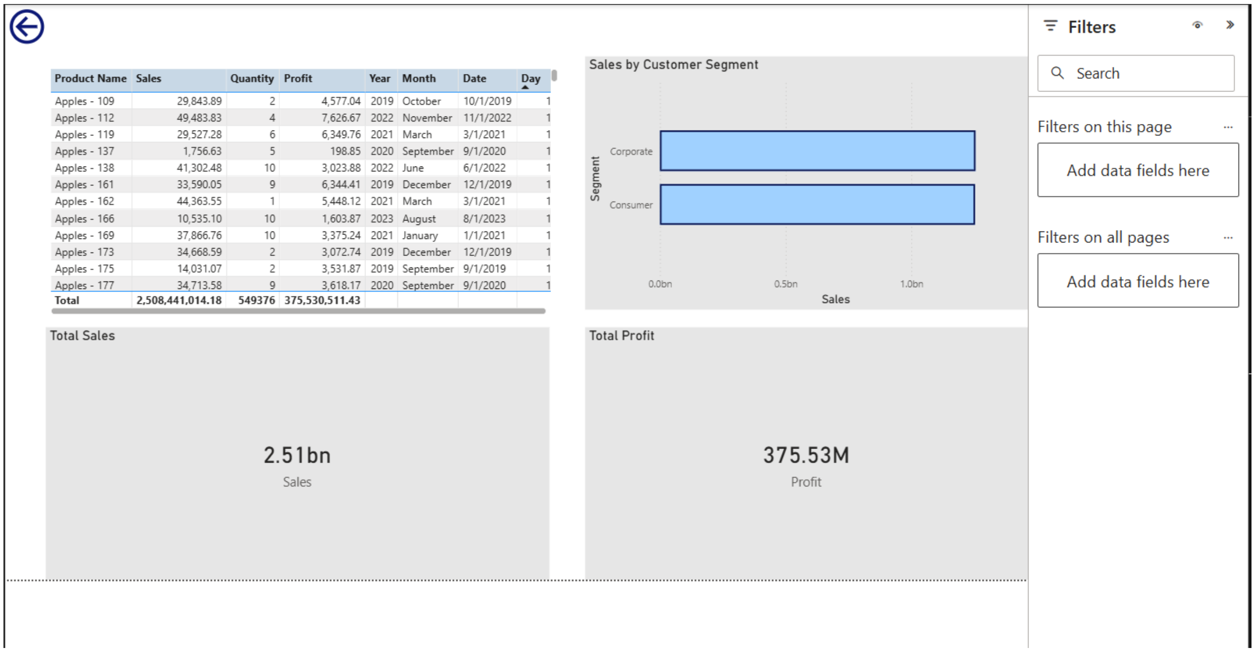
Goal:

Create a report that enables users to right-click on a data point and navigate to a detailed page showing related information.

Steps:

1. Created a new report page named **"Drill-Through"**.
2. Added a **Drill-through filter** by dragging **Product Name (from DimProduct)** into the Drill-through filters section.
3. Designed the drill-through page by adding detailed visuals:
   - A **Table** showing Product Name, Sales, Quantity, and Profit.
   - **Card visuals** displaying Total Sales and Total Profit.
4. Inserted a **Back button** to allow users to return to the main report page.
5. On the main report page, created a visual (e.g., Column Chart) with Product Category and Sales.
6. Tested the drill-through functionality:

- Right-clicked on a category in the chart.
- Selected **Drill Through → Drill-Through Report**.
- Verified that the page displays filtered data relevant to the selected item.



Product Name	Sales	Quantity	Profit	Year	Month	Date	Day
Apples - 109	29,843.89	2	4,577.04	2019	October	10/1/2019	1
Apples - 112	49,483.83	4	7,626.67	2022	November	11/1/2022	1
Apples - 119	29,527.28	6	6,349.76	2021	March	3/1/2021	1
Apples - 137	1,756.63	5	198.85	2020	September	9/1/2020	1
Apples - 138	41,302.48	10	3,023.88	2022	June	6/1/2022	1
Apples - 161	33,590.05	9	6,344.41	2019	December	12/1/2019	1
Apples - 162	44,363.55	1	5,448.12	2021	March	3/1/2021	1
Apples - 166	10,535.10	10	1,603.87	2023	August	8/1/2023	1
Apples - 169	37,866.76	10	3,375.24	2021	January	1/1/2021	1
Apples - 173	34,668.59	2	3,072.74	2019	December	12/1/2019	1
Apples - 175	14,031.07	2	3,531.87	2019	September	9/1/2019	1
Apples - 177	34,713.58	9	3,618.17	2020	September	9/1/2020	1
Total	**2,508,441,014.18**	**549376**	**375,530,511.43**				